Stochastic Processes
Spring 2025 March 6, 2025

Homework #3: Discrete Time Markov Chain
Due: March 20th, noon

#1 Consider an irreducible and recurrent Markov chain. Argue that, starting from any state, the chain will visit every other state with probability 1 at some future time.

#2 Consider a warehouse with an initial inventory level of 0. Time is discrete. At the beginning of each day, with probability $\lambda \in (0,1)$, one unit of inventory arrives at the warehouse. At midday, with probability $\mu \in (0,1)$, there is a demand for one unit of inventory. If the demand occurs and there is available inventory, one unit is used to fulfill the demand (each unit of inventory can only be used once). Unfilled demand is lost. Let X_n be the number of unused inventory units at the end of day n. Let Y_m be the number of unused inventory units before the m-th change of the inventory level. For example, when $X_0 = 0, X_1 = 1, X_2 = 1$ and $X_3 = 0$, then $Y_0 = 0, Y_1 = 1, Y_2 = 0$.

- Argue why $\{X_n, n \geq 0\}$ is an irreducible DTMC. Show the transition matrix.
- Argue why $\{Y_m, m \geq 0\}$ is an irreducible DTMC. Show the transition matrix.
- For $\{Y_m, m \geq 0\}$, under what conditions state i is transient or recurrent? **Hint**: (1) Starting from 0, the next state will always be 1. (2) Starting from 1, the probability of hitting 0 is similar to that in the classical gambler's ruin model.
- For $\{X_n, n \geq 0\}$, under what conditions state i is transient or recurrent? Hint: connect it to the previous question.
- If $\{X_n, n \geq 0\}$ is positive recurrent, show its long-run proportion.

#3 A factory operates a circular production system with $n+1$ workstations arranged in a loop, labeled as $0, 1, \ldots, n$. A quality inspection robot moves between these stations to ensure that each workstation is functioning correctly. The robot starts its inspection at workstation 0. During its movement, the robot follows specific transition rules. When at workstation i ($0 \leq i \leq n$), it has a probability $q_i \in (0,1)$ of remaining at the same station for the next inspection. With probability $q_i p$, it moves clockwise to station $(i+1) \mod (n+1)$, and with probability $q_i(1-p)$, it moves counterclockwise to station $(i-1) \mod (n+1)$. Given these movement rules,

- What is the probability that the robot inspects workstation $i \in \{1, \cdots, n\}$ at least once before returning to station 0 for the first time? **Hint**: After the first move, connect the problem to the gambling ruin model, where the two absorbing states are 0 and i.

- What is the probability that the robot inspects all workstations at least once before returning to station 0 for the first time?
- Suppose the robot inspects all workstations at least once before returning to station 0 for the first time. Let T_k be the first time inspecting station k. What's the probability that $T_1 < T_2 < \cdots < T_n$.

#4 **(Chapter 4, Exercise 64)** Consider a branching process having $\mu < 1$. If $X_0 = n$, calculate the expected number of individuals that ever exist in this population.

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